

UNIFIED MENTOR

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Batch	25Mar thru 25Jun 2026
Type of Intern	Data Analytics Internship – HR Analytics
Project Title	2. Employee Engagement, Satisfaction, and Burnout Diagnostic Analysis at Palo Alto Networks
Github Repo Link	https://github.com/M339KUMAR/DataAnalytics2/
Project Pdf Link	
Video Link	https://github.com/M339KUMAR/DataAnalytics2/blob/main/P2%20-%20Human%20Resources%2020260607_145551.mp4
Streamlit Project Link	https://dataanalytics2-umdap2hraproject2.streamlit.app/

Abstract :-

The Human Resources Department @Palo Alto Networks Company has been facing the problem of Employee Attrition and is unable to identify the factors driving the same & also the Risk assessment of Employees leaving the company, So the HR department was reactive rather than Proactive, So the Dept decided build a new system that completely aid them in workforce planning & management. The present study using Data Analysis using Past historical data aims design & build a new system which can help the Manager of HR Dept to plan ahead & take necessary steps of identifying which particular employees in which dept are more risk prone to leaving, & accordingly take necessary steps to maintain Harmony in the Company.

Problem Definition:-

In The present Existing system HR leaders are facing trouble to correctly identify the factors behind employees leaving the company so they try to use the past historical data to design a predictive intelligence that enables Early Identification, Targeted Retention & Data-Driven workforce planning.

The Dataset Description :-

The Dataset contains 31 variables with Attrition as Target Variable(included) and 1470 records

Objectives:-

- Identification of High Risk employees
- Targeted Retention Interventions
- Data Driven workforce planning
- Loss of high performing or critical employees
- Visibility into key drivers influencing exit
- A Systematic way to predict attrition.

Methodology :-

- Data collection
- Data Preprocessing
- Feature Engineering
- ML Model Development
- Risk Scoring Framework

System Design :-

The new system identifies the problematic areas by building a Dashboard using the streamlit, other libraries using the python Programming language The system build a Machine Learning model using Logistic Regression, Random Forest & Xgboost Models, with Log Reg Model as a Baseline Model, The key Performance metrics of RF model are also Displayed and modelcomparison chart is displayed to analyse which model best suits our needs, accordingly HR Manager can plan the workforce of the company.

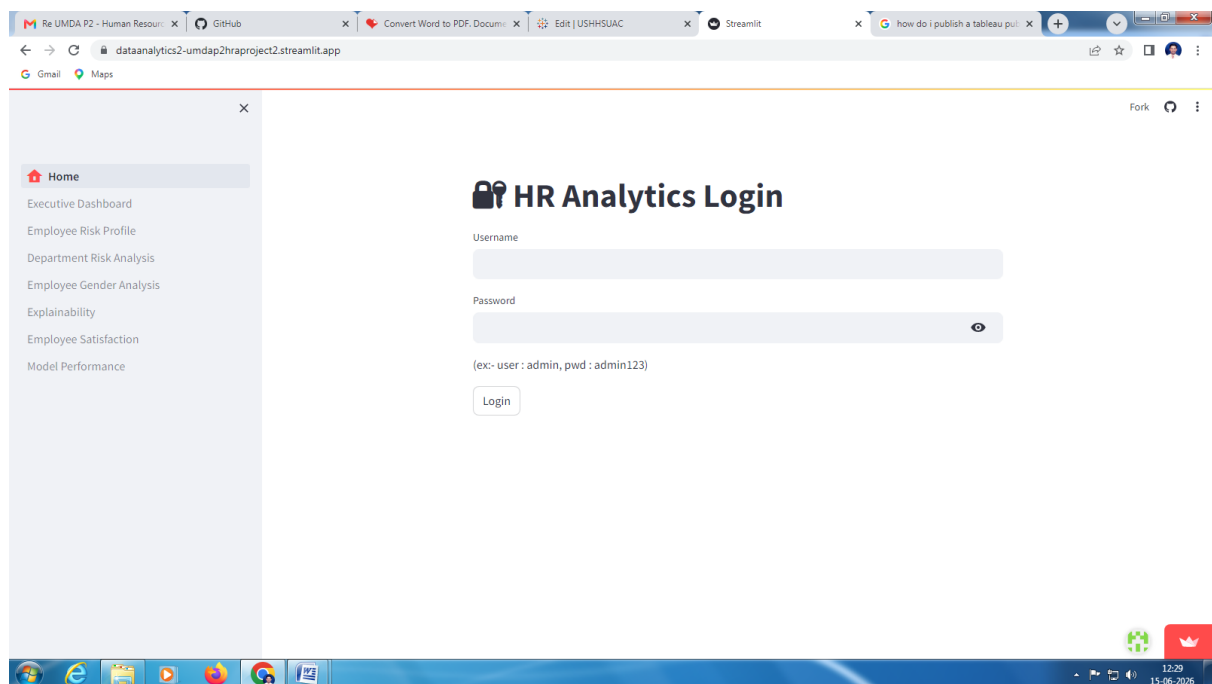
Conclusion & Recommendation: -

The present report makes an attempt to aid the HR Manager but does not replace the same, no matter how good the model is the human judgement always dominates the ML Model, so care should be taken while using the ML model, with this context in mind, This New system design transforms the HR Analytics @Po Alto Networks from descriptive & diagnostic insights into predictive decision intelligence, by forecasting employee attrition risk & providing the explainability insights the organizationcan move from reactive retention effort to proactive, employee-cetric workforce Management.

(In order to make more visually appealing and understandable screenshots are also attached, the user can first visit the home page and login using the credentials , for ex:- username : admin & password : admin123).

Home Page - Auth.py (Login Authorization)

User ID	Password
admin	admin123
hrmanager	hr2026

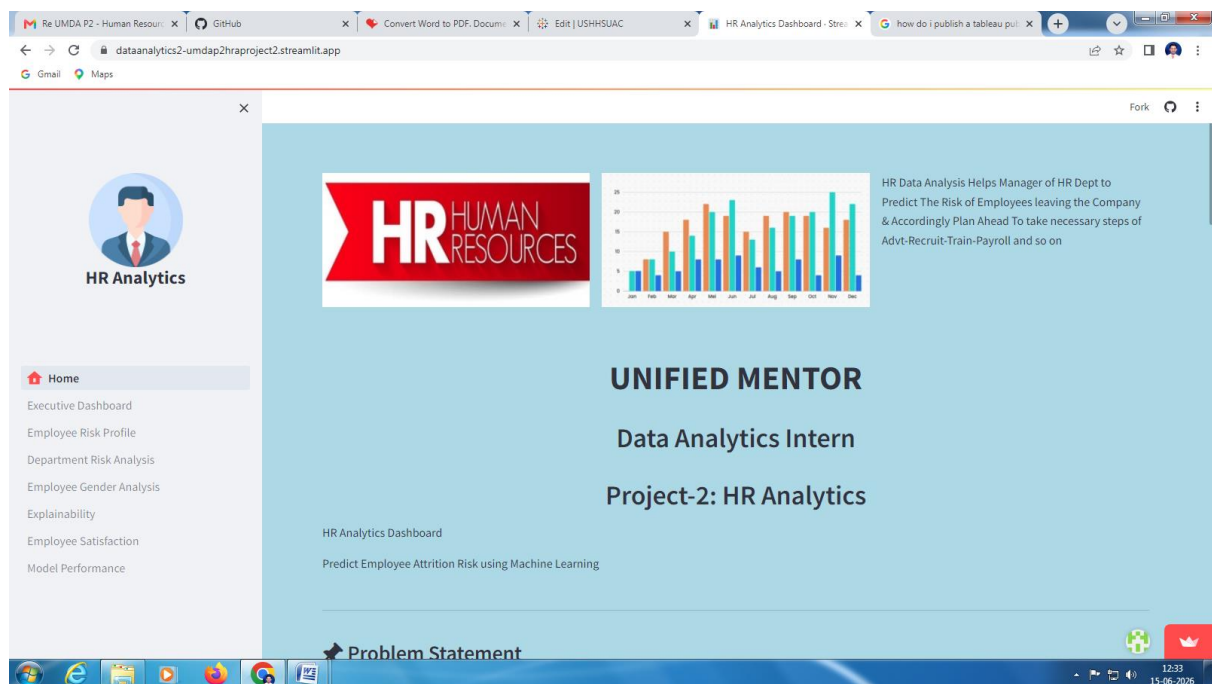


After Login The Side bar Navigation Panel Displays List of Pages in HR Analytics Dashboard.

1. Executive Dashboard
2. Employee Risk Profile
3. Department Risk analysis
4. Employee Gender Analysis
5. Explainability
6. Employee Satisfaction
7. Model Performance

The Home Page gives Introduction of :

- Problem Statement
- Organization Need
- Methodology
- Dataset Information



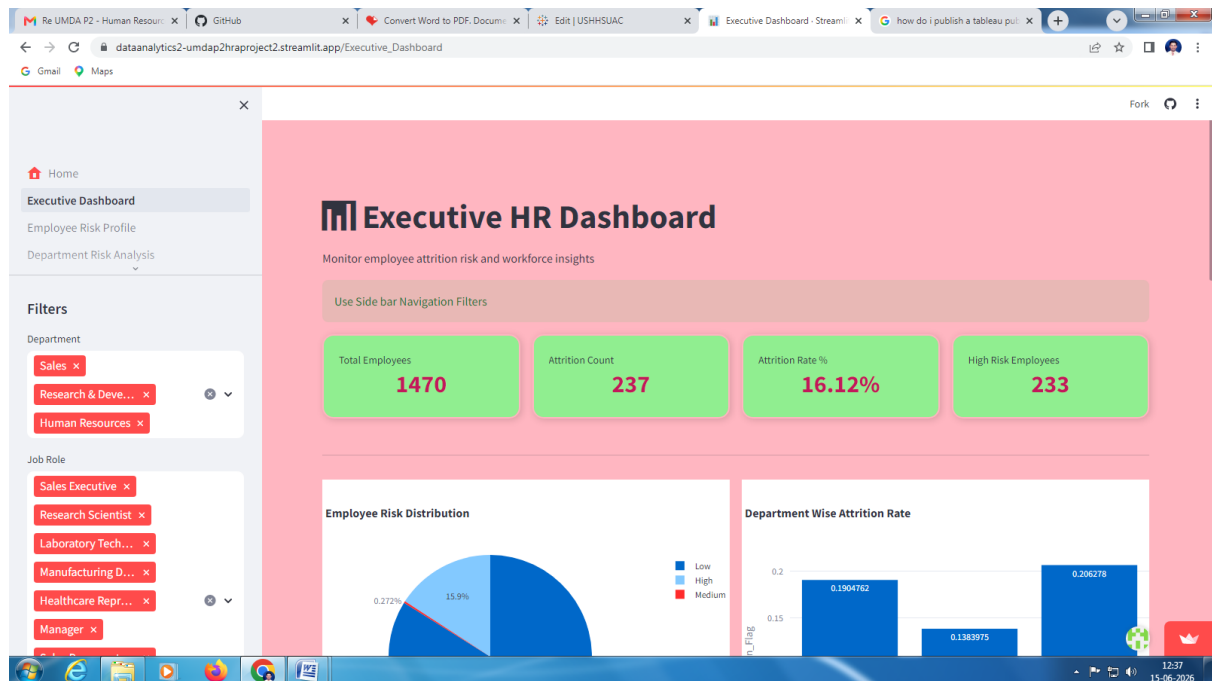
1st Page is - Executive Dashboard

KPI in Page-1 are:

Total Employees in Company
Attrition Count
Attrition Rate
High Risk Employees

Charts & Graphs:

Employee Risk Distribution
Dept Wise Attrition Rate
Risk by Job Role
Income vs Overtime
Dataset Showing High Risk Employees



2nd Page is - Employee Risk Profile

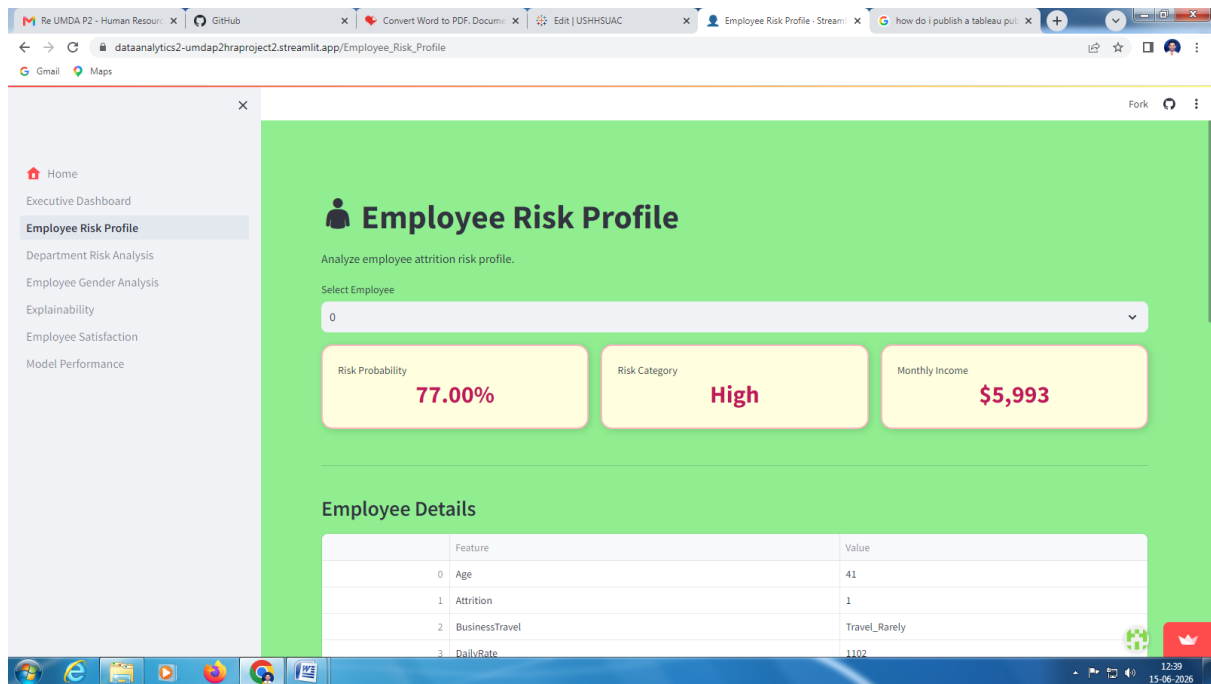
KPI in Page-2 are:

Risk Probability,
Risk Category,
Monthly Income,
Example Emp Details.

Charts & Graphs:

Top Risk Drivers

Employee Risk Score of Attrition



3rd Page is - Department Risk Analysis

KPI in Page-3 are:

High Risk Dept

High Attrition Dept

Average Risk %

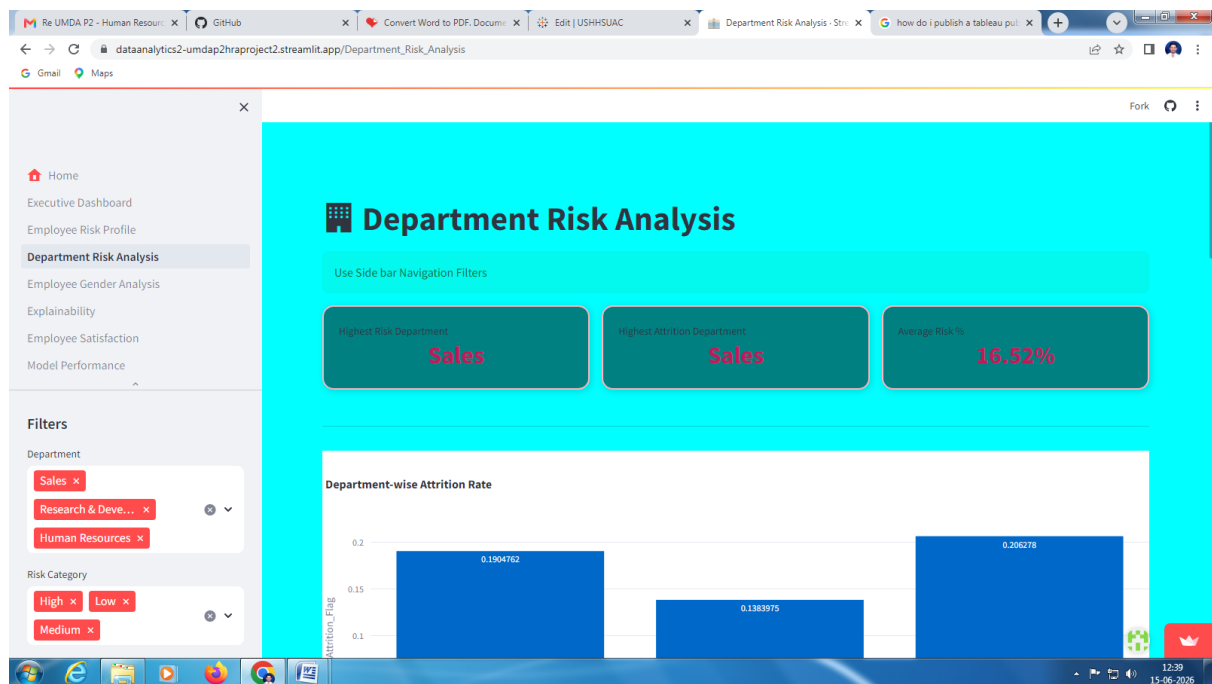
Charts & Graphs:

Dept wise Attrition Rate

Avg Risk Score By Dept

Job Role Risk Ranking

Risk Dist across Dept



4th Page is - Gender Analysis of Employees

KPI in Page-4 are:

Total Employees

Male Employees

Female Employees

Charts & Graphs:

There are 3 tabs :

Tab-1: Dataset Explorer,

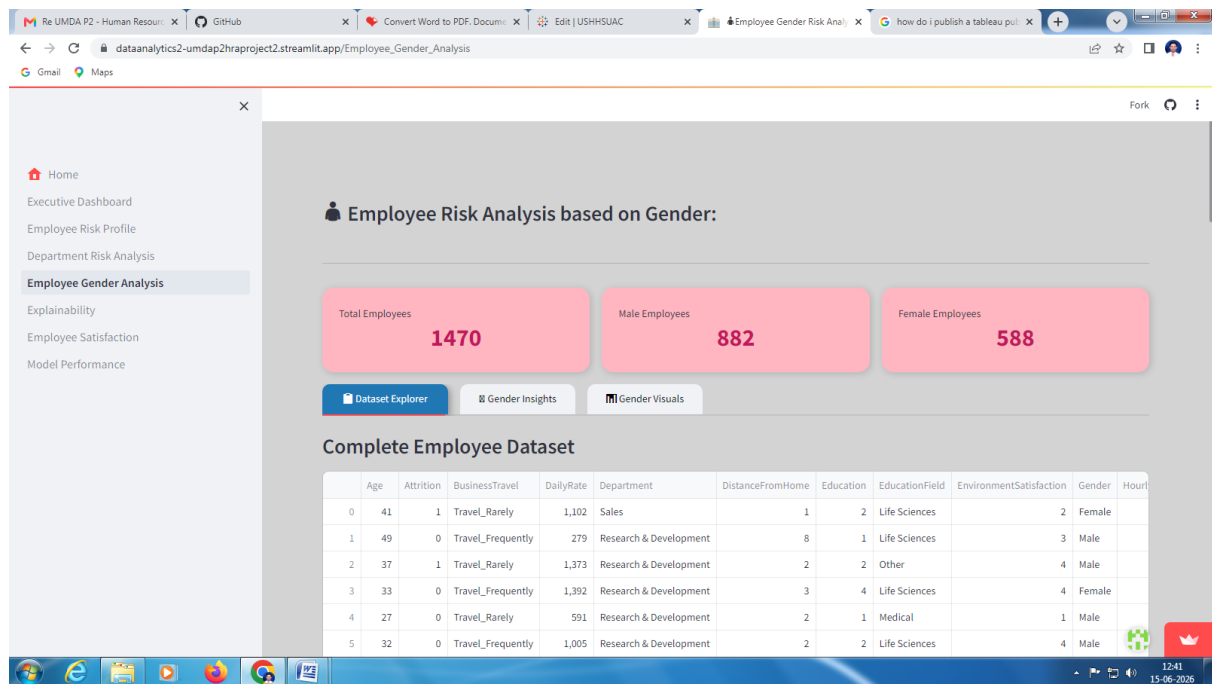
Tab-2: Gender Insights,

Tab-3: Gender Visuals.

Male Employees across Dept

Female Employees across Dept

Dept wise Male-Female count



5th Page - Employee Attrition Explainability

KPI in Page-5 are:

Risk Score

Risk Category

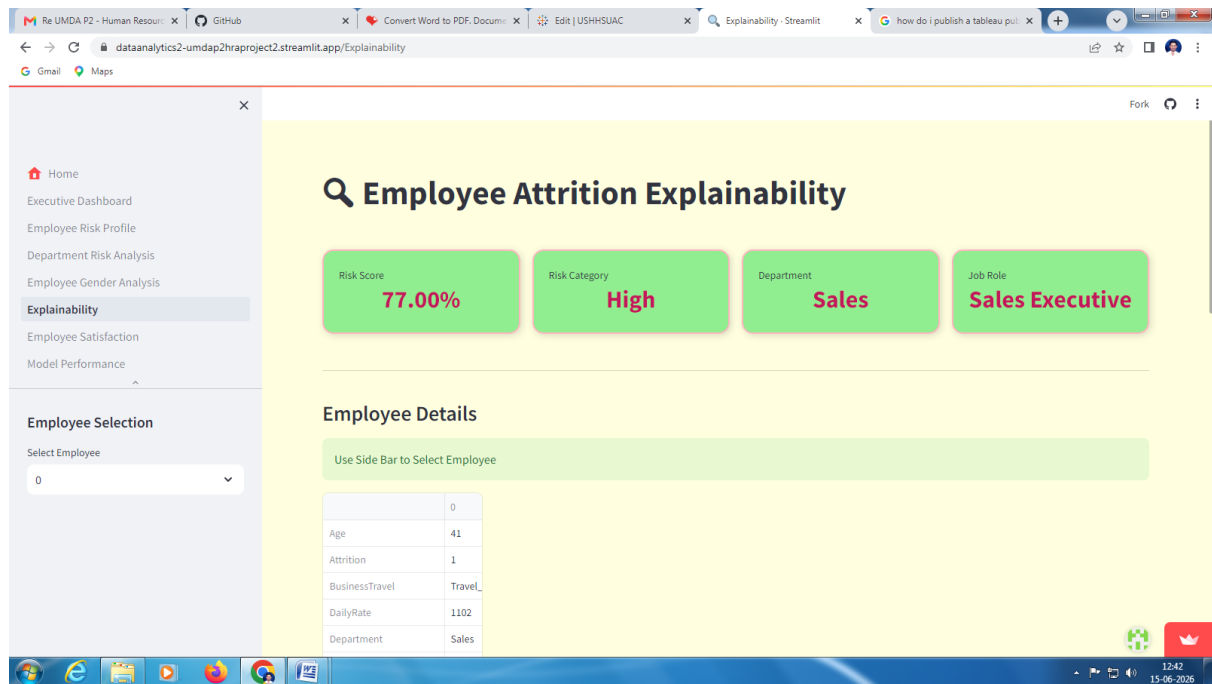
Department

Job Role

Charts & Graphs:

Employee Details of User Selection

Top Factors Driving the Attrition



6th Page is - Employee Satisfaction Stats;

KPI in Page-6 are:

Job Satisfaction Rating
Envi Satisfaction Rating
Relationship Satisfaction Rating
Work-Life Balance Score

Charts & Graphs:

there 3 Tabs here

Tab-1: Dept wise Job Satisfaction

Tab-2: Dept wise Envi Satisfaction

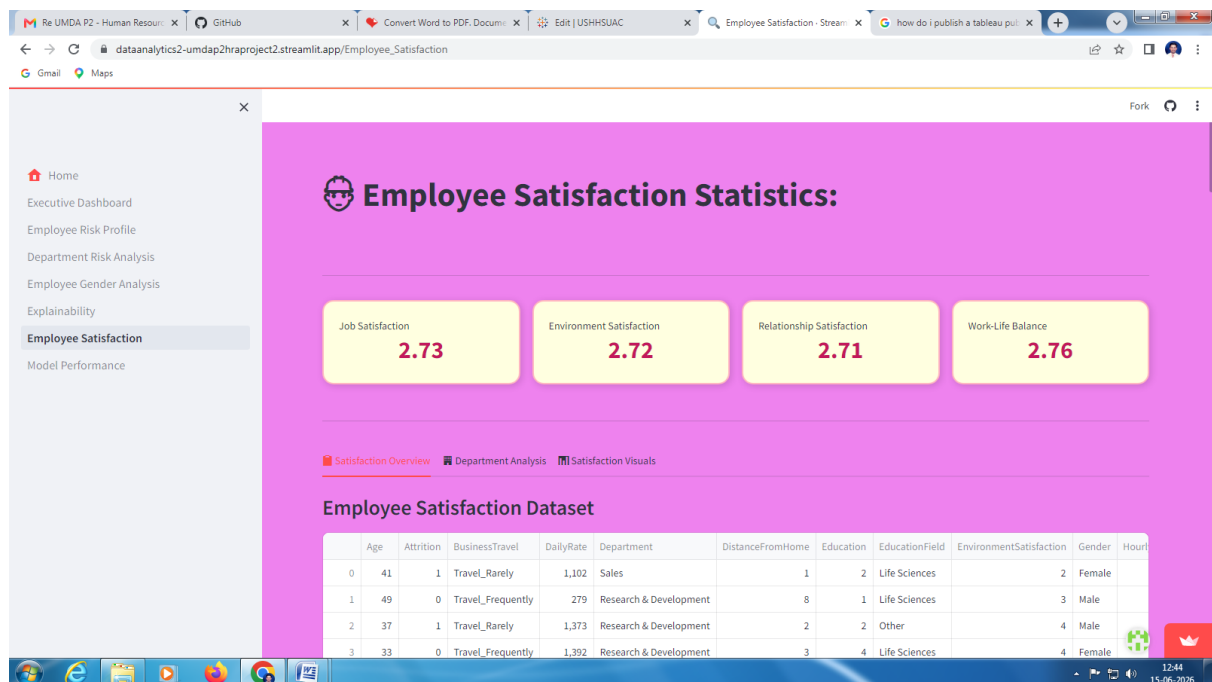
Tab-3: Dept wise Work-Life Balance

Top 10 Most Satisfied Employees Bottom 10 Least Satisfied Employees

Attrition Based on Marital Status

Attrition Dist of Male & Female Employees based on their Marital Status.

Finally **Overall Employee Satisfaction** of the company across All Dept in the form of Rating. (2.73 on 4 point scale)



7th Page is - Model Evaluation

KPI in Page-7 are:

Dataset Overview

Total Variables

Target Variable

Charts & Graphs:

Attrition Class Distribution

SMOTE Balancing

Model Training

Model Comparison Table:

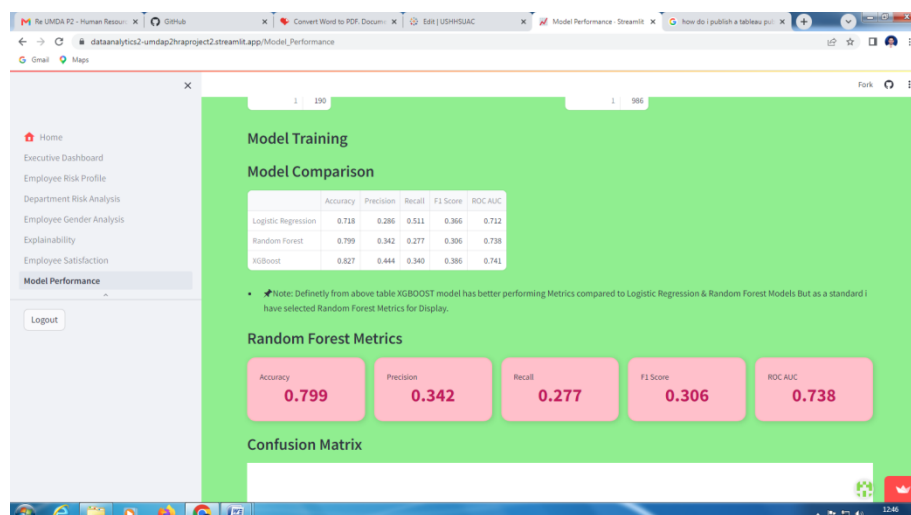
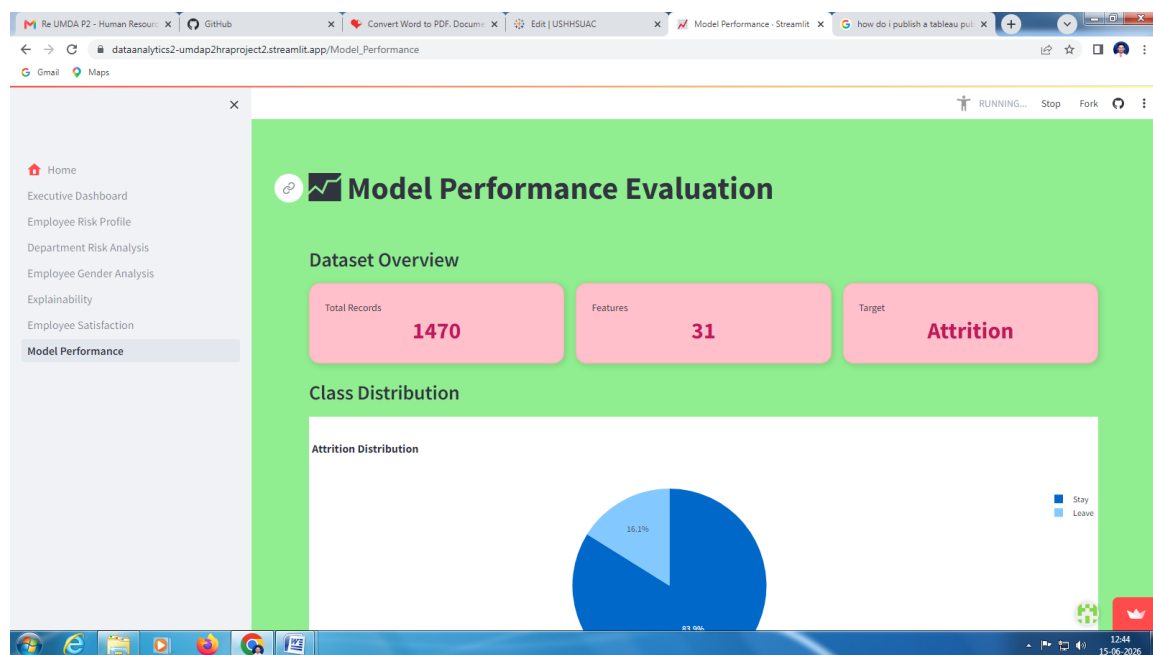
(Baseline-Logistic Regression)

(Adv- Random Forest, XGBoost)

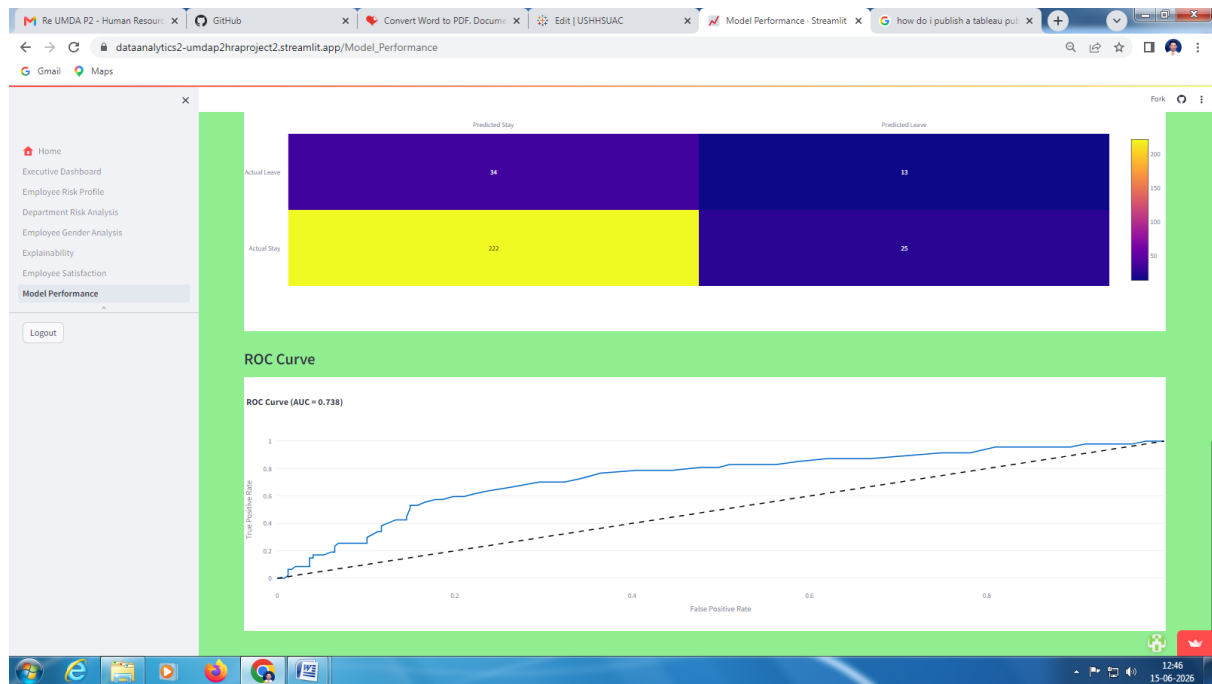
Random Forest Metrics

Confusion Matrix

ROC-AUC Curve



Finally User can Logout from the Dashboard from the Side bar Navigation Panel using a Logout Button in the Model Performance Page.



Logout Button available in the sidebar Navigation

THANK YOU